DEPARTEMENT STATISTIEK EN AKTUARIËLE WETENSKAP
DEPARTEMENT OF STATISTICS AND ACTUARIAL SCIENCE

STATISTIEK EN DATAWETENSKAP 188
STATISTICS AND DATA SCIENCE 188

TUTORIAAL 11 / *TUTORIAL* 11

Die meeste probleme is aangepas uit BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall. / *Most problems are adapted from BASIC Business Statistics – Berenson, Levine & Krebiel, 10th ed, Prentice Hall*.

1. Beskou die Excel datastel "regression_data.xlsx" en beantwoord die vrae wat volg: / *Consider the Excel dataset called "regression_data.xlsx" and answer the questions that follow:*
   (a) Gebruik die "Data Analysis Tool" om die y-afsnit en helling van die regressielyn tussen X en Y te beraam. / *Use the Data Analysis Tool to estimate the intercept and slope of the regression line between X and Y.*
   (b) Selekteer alle uitsette van die "Data Analysis Tool" vir regressie en ondersoek die verskillende statistieke en grafieke wat Excel lewer. / *Select all outputs from the Data Analysis Tool for regression and investigate the different statistics and graphs that Excel produce.*

2. 'n Reguit lyn word op 'n datastel gepas en die volgende regressielyn word verkry: / *Fitting a straight line to a set of data yields the following regression line:*

$$\widehat{Y}_i = 2 + 5X_i$$

   (a) Interpreteer die betekenis van die Y-afsnit, b_0. / *Interpret the meaning of the Y intercept b_0.*
   (b) Interpreteer die betekenis van die helling, b_1. / *Interpret the meaning of the slope b_1.*
   (c) Voorspel die gemiddelde waarde van Y vir $X = 3$. / *Predict the mean value of Y for $X = 3$.*

3. 'n Reguit lyn word op 'n data stel gepas en die volgende regressielyn word verkry: / *Fitting a straight line to a set of data yields the following regression line:*

$$\widehat{Y}_i = 16 - 0.5X_i$$

   (a) Interpreteer die betekenis van die Y-afsnit, b_0. / *Interpret the meaning of the Y intercept b_0.*
   (b) Interpreteer die betekenis van die helling, b_1. / *Interpret the meaning of the slope b_1.*
   (c) Voorspel die gemiddelde waarde van Y vir $X = 6$. / *Predict the mean value of Y for $X = 6$.*

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4. Die bemarkingsbestuurder van ’n groot supermark wil die benutting van rakspasie gebruik om die verkope van troeteldierkos te voorspel. ’n Ewekansige steekproef van twaalf ewe groot winkels wat troeteldierkos verkoop is gekies, met die volgende resultate: / *The marketing manager of a large supermarket chain would like to use shelf space to predict the sales of pet food. A random sample of twelve equal-sized stores selling pet food is selected, with the following results*:

Winkel / *Store*	Rakspasie (in meter) / *Shelf space (in meter)* ***(X)***	Weeklikse verkope (in honderde rand) / *Weekly sales (in hundreds of rand)* ***(Y)***
1	5	1.6
2	5	2.2
3	5	1.4
4	10	1.9
5	10	2.4
6	10	2.6
7	15	2.3
8	15	2.7
9	15	2.8
10	20	2.6
11	20	2.9
12	20	3.1

Die data lewer $b_0 = 1.45$ en $b_1 = 0.074$. / *The data yields* $b_0 = 1.45$ *and* $b_1 = 0.074$.

(a) Interpreteer die betekenis van die helling, b_1 in hierdie probleem. / *Interpret the meaning of the slope* b_1 *in this problem*.

(b) Voorspel die gemiddelde weeklikse verkope (in honderde rande) van winkels wat 8 meter rakspasie aan troeteldier kos afstaan. / *Predict the mean weekly sales (in hundreds of rands) of pet food for stores with* 8 *meters of shelf space for pet food*.

5. In Probleem 4 het die bemarkingsbestuurder rakspasie gebruik om gemiddelde weeklikse verkope mee te voorspel. Die data lewer *SSR* = 2.0535 en *SST* = 3.0025. / *In Problem 4 the marketing manager used shelf space to predict mean weekly sales. The data yields SSR* = 2.0535 *and SST* = 3.0025.

(a) Bereken die bepaaldheidskoëffisiënt, r^2 en interpreteer die betekenis daarvan. / *Determine the coefficient of determination* r^2 *and interpret its meaning.*

(b) Bereken die standaardfout van die beraming. / *Determine the standard error of estimation.*

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(c) Hoe nuttig dink u is hierdie regressiemodel om verkope mee te voorspel? / *How useful do you think this regression model is for predicting sales*?

6. Indien *SSR* = 36 en *SSE* = 4, bereken *SST*, en bereken ook die bepaaldheidskoëffisiënt, r^2 en interpreteer die betekenis daarvan. / *If SSR* = 36 *and SSE* = 4, *calculate SST, and then compute the coefficient of determination,* r^2 *and interpret its meaning*.

7. Indien *SSR* = 66 en *SST* = 88, bereken die bepaalheidskoëffisiënt, r^2 en interpreteer die betekenis daarvan. / *If SSR* = 66 *and SST* = 88, *compute the coefficient of determination,* r^2 *and interpret its meaning*.

8. Indien *SSE* = 10 en *SSR* = 30, bereken die bepaalheidskoëffisiënt, r^2 en interpreteer die betekenis daarvan. / *If SSE* = 10 *and SSR* = 30, *compute the coefficient of determination,* r^2 *and interpret its meaning*.

9. Veronderstel u toets die nulhipotese dat daar geen verband is tussen die twee veranderlikes, *X* en *Y* nie. Die steekproefgrootte is $n = 18$, $b_1 = 4.5$ en $s_{b_1} = 1.5$. / *You are testing the null hypothesis that there is no relationship between two variables X and Y. From your sample of* $n = 18$, $b_1 = 4.5$ *and* $s_{b_1} = 1.5$.

   (a) Wat is die waarde van die *t*-toetsstatistiek? / *What is the value of the t-test statistic*?
   (b) By die $\alpha = 0.05$ betekenispeil, wat is die kritieke waardes? / *At the* $\alpha = 0.05$ *level of significance, what are the critical values*?
   (c) Gebaseer op u antwoorde in (a) en (b), watter statistiese beslissing sal u maak? / *Based on your answers to (a) and (b), what statistical decision should you make*?
   (d) Konstrueer 'n 95% vertrouensintervalberaming vir die helling in die model vir die populasie. / *Construct a* 95% *confidence interval estimate for the slope in the model for the population*.

10. Verwys na Probleem 4. Die data in hierdie problem lewer $b_1 = 0.074$ en $s_{b_1} = 0.0159$. / *Refer Problem 4. The data in this problem yields* $b_1 = 0.074$ *and* $s_{b_1} = 0.0159$.

    (a) Is daar bewys van 'n lineêre verwantskap tussen rakspasie en verkope by die 0.05 betekenispeil? / *At the* 0.05 *level of significance, is there evidence of a linear relationship between shelf space and sales*?
    (b) Konstrueer 'n 95% vertrouensintervalberaming vir die helling van die model vir die populasie. / *Construct a* 95% *confidence interval estimate for the slope of the model for the population*.

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11. Die data op die aangehegte bladsy is gebaseer op inligting ten opsigte van ’n baie bekende ramp in die geskiedenis, die sg. Challenger ramp. Die rampspoedige lansering het by ’n temperatuur van 4 °C plaasgevind. Die data behels temperatuur teenoor O-Ring skade indeks. (Let op dat die werklike verband nie lineêr is nie, maar wel nie-lineêr. Die lineêre benadering gee egter ’n goeie idee van moontlike afhanklikheid. Indien u statistiek verder sou bestudeer, sal u sien hoe hierdie probleem op ’n baie elegante wyse benader word.) / *The data on the page attached is based on the information on a well-known disaster in history, the so-called Challenger disaster. The disastrous launch took place at a temperature of* 4 *°C. The data consider temperature vs O-Ring damage index. (Note that the actual relationship is not linear, but non-linear. The linear approach, however, does give a fairly good idea of a possible dependency. If you continue your study of statistics, you will see how this regression problem is solved in a very elegant way.)*

Bestudeer die afvoer en voltooi die volgende: / *Consider the output and complete the following:*

(a) Die beraamde regressievergelyking (model) is: / *The estimated regression equation (model) is:*

(b) Die aantal vryheidsgrade vir die residu [*df(residu)*] is*: / The number of degrees of freedom for the residual [df(residual)] is:*

(c) *SSE* =

(d) Die gemiddelde som van kwadrate vir residue / *The mean sum of squares of residuals*

$$\frac{\sum(y-\hat{y})^2}{n-2}$$

(e) Standaardfout van die beraming / *Standard error of the estimate* =

(f) $F_b = F_{calc}$ =

(g) r^2 = SSR/SST =

(h) $100r^2$ =_____, en word geïnterpreteer as _____./ *and is interpreted as* _____.

(i) Die 95% vertrouensintervalberaming vir β_1 = _____. / *The* 95*% confidence interval estimate for* β_1 = _____.

(j) Gee drie maniere waarop die volgende hipoteses getoets kan word m.b.v. die inligting in die afvoer tesame met van u vorige resultate: / *Give three ways in which the following hypotheses can be tested using the information in the output together with your previous results*:

$H_0 : \beta_1 = 0$

$H_1 : \beta_1 \neq 0$

$\alpha = 0.05$

(k) Beraam die O-Ring skade indeks by ’n temperatuur van 11.67 °C. / *Estimate the O-Ring damage index at* 11.67 *°C.*

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(l) Bereken die ooreenkomstige residu. */ Calculate the corresponding residual.*

SUMMARY OUTPUT

	Regression Statistics
Multiple R	0.5264
R Square	**(g)**
Adj R Square	0.2443
Stand Error	**(e)**
Observations	24

ANOVA

	df	*SS*	*MS*	*F*	*Sign F*
Regression	1	3.8339	3.8339	**(f)**	0.0082
Residual	**(b)**	**(c)**	**(d)**		
Total	23	13.8333			

	Coeff	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	2.5378	0.7432	3.4147	0.0025	0.9965	4.0791
Launch Temp °C (X)	-0.1003	0.0345	-2.9043	0.0082	**(i)**	**(i)**

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RESIDUAL OUTPUT

Observation	*Launch Temp °C (X)*	*O-Ring Damage Index (Y)*	***Predicted** O-Ring Damage Index (Y)*	*Residuals*
1	11.67	3	**(k)**	**(l)**
2	13.33	1	1.2011	-0.2011
3	13.89	1	1.1454	-0.1454
4	17.22	0	0.8112	-0.8112
5	19.44	0	0.5884	-0.5884
6	19.44	0	0.5884	-0.5884
7	19.44	0	0.5884	-0.5884
8	20.00	0	0.5327	-0.5327
9	20.56	0	0.4770	-0.4770
10	21.11	0	0.4213	-0.4213
11	21.11	0	0.4213	-0.4213
12	21.11	1	0.4213	0.5787
13	21.11	1	0.4213	0.5787
14	21.11	1	0.4213	0.5787
15	22.22	0	0.3099	-0.3099
16	22.78	0	0.2542	-0.2542
17	23.89	0	0.1428	-0.1428
18	23.89	2	0.1428	1.8572
19	24.44	0	0.0871	-0.0871
20	24.44	0	0.0871	-0.0871
21	25.56	0	-0.0243	0.0243
22	26.11	0	-0.0800	0.0800
23	26.67	0	-0.1357	0.1357
24	27.22	0	-0.1914	0.1914

12. ‘n Meervoudige lineêre regressiemodel tussen *Y* en nege onafhanklike veranderlikes word beraam vanaf ‘n steekproef van grootte 64. Die model lewer ‘n SST = 53895.51 en ‘n MSR = 1455. / *A multiple linear regression model between Y and nine independent variables is estimated from a sample of size 64. The model gives an SST = 53895.51 and an MSR = 1455.*
    (a) Bereken die SSR en SSE. / *Calculate the SSR and SSE.*
    (b) Bereken die MSR en MSE. / *Calculate the MSR and MSE.*
    (c) Bereken die steekproefstandaardafwyking van *Y*. / *Calculate the sample standard deviation of Y.*
    (d) Bereken die bepaalheidskoëffisiënt, r^2, en interpreteer die betekenis daarvan. / *Compute the coefficient of determination, r^2, and interpret its meaning.*
    (e) Bereken die korrelasiekoëffisiënt. / *Calculate the coefficient of correlation.*
    (f) Bereken die aangepaste r^2. / *Calculate the adjusted r^2.*

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(g) Definieer die nul- en alternatiewe hipotese om te toets vir algehele betekenisvolheid. / *Define the null and alternative hypothesis to test for the overall significance of the model.*

(h) Wat is die waarde van die F-toetssatistiek in vraag (g)? / *What is the value of the F-test statistic in question (g)?*

(i) Indien u hierdie hipotese by die 5% betekenispeil toets, wat is u beslissing? / *If you test this hypothesis at the 5% level of significance, what is your conclusion?*

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